Introduction

Along these tutorials, we will see how **SCOPE** interacts with the different parts of the code to handle the execution of computational workflows.

These are the topics covered in each tutorial:

1. The **System** class and its sources: the **Specie**, **Cell** and **Atom** classes
2. The Computational workflow: **Branch**, **Workflow**, **Job**, and **Computation** classes
3. The **State** class
4. The **Data**, **Collection** and **VNM** classes
5. The **Input_data** class, and **scope input files**
6. Running **SCOPE** - Part 1: File Structure
7. Running **SCOPE** - Part 2: Execution
8. Running **SCOPE** - Part 3: Detailed Actions
9. The Azo add-on: Subclasses, Analysis and Creation of Azo Systems

Tutorial 3: The State class

In this tutorial, we will discuss the **State** class. A **State** is a 'picture' of a given **Source** during the execution of the computational Workflow. The main purpose of a **State** is to store the initial data for **Computations**, which have the ability to create a new state upon completion, or to update an existing one, depending on the user choices.

Below, we will discuss how the **State** function works exactly, and how it connects **Species** (and **Cells**) with the computational workflow.

In [1]:
```
import os
import scope
import numpy as np
```

In [2]:
```
## Path of the data folder. It should be "os.path.abspath('.')+'/Data"
data_folder = os.path.abspath('../')+'/Data/1-Tutorial_1/'

## Loads the System object from a binary file, provided in the tutorial folc
sys = scope.load_binary(f"{data_folder}/Systems/ABITEM/ABITEM.npy")
```

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```
## This will set the main path of the system to the data folder, so that any
sys.set_main_path(data_folder)
```

Out[2]: True

In [3]:
```
## All objects in SCOPE have a __repr__ method, so printing shows a summary
print(sys)
```

```
-------------------------------------
-- >>> SCOPE System_sco Object >>> --
-------------------------------------
 Version                = 1.0
 Type                   = system
 Subtype                = sco_system
 Source Path            = /Users/sergivela/Documents/SCOPE/Program/Scope_Tuto
rials/Data/1-Tutorial_1/Sources/ABITEM/
 System File Path       = /Users/sergivela/Documents/SCOPE/Program/Scope_Tuto
rials/Data/1-Tutorial_1/Systems/ABITEM/
 System File Name       = /Users/sergivela/Documents/SCOPE/Program/Scope_Tuto
rials/Data/1-Tutorial_1/Systems/ABITEM/ABITEM.npy
 Computations Path      = /Users/sergivela/Documents/SCOPE/Program/Scope_Tuto
rials/Data/1-Tutorial_1/Computations/ABITEM/

 Num Sources            = 6
     idx: type, name, formula
     0: cell, ABITEM01, H104-C92-N24-O4-S8-Fe4
     1: cell, ABITEM, H104-C92-N24-O4-S8-Fe4
     2: cell, ref_hs_cell, H104-C92-N24-O4-S8-Fe4
     3: cell, ref_ls_cell, H104-C92-N24-O4-S8-Fe4
     4: specie, ref_hs_mol, H18-C20-N6-S2-Fe
     5: specie, ref_ls_mol, H18-C20-N6-S2-Fe

 Num Branches           = 1
     0: isolated, Num Workflows: 2
```

In [4]:
```
## Sources can be selected using the find_source function, and using the sou
found, source = sys.find_source("ref_hs_mol")
print(source)
```

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```
-----------------------------------
------ SCOPE MOLECULE Object ------
-----------------------------------
 Version               = 1.0
 Type                  = specie
 Sub-Type              = molecule
 Name                  = ref_hs_mol
 Number of Atoms       = 47
 Formula               = H18-C20-N6-S2-Fe
 Charge                = 0
 Spin (alpha - beta)   = 4
 Number of Parents     = 1
 Has Adjacency Matrix  = YES
 Has Bonds             = YES

 Num of Ligands        = 3
   Ligand 0: C-N-S with 3 atoms. Smiles: [N-]=C=S
   Ligand 1: H18-C18-N4 with 40 atoms. Smiles: [H]c1nc(C([H])([H])N(C([H])
([H])c2nc([H])c([H])c([H])c2[H])C([H])([H])c2nc([H])c([H])c([H])c2[H])c([H])
c([H])c1[H]
   Ligand 2: C-N-S with 3 atoms. Smiles: [N-]=C=S

 Num of Metals         = 1
   Metal 0: Fe with charge:2 and spin:4
```

States

In [5]:
```
## Sources have "STATES"
source.states
```

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Out[5]:
```
[--------------------------------
------ SCOPE STATE Object ------
--------------------------------
 Name                  = initial
 Source Name           = ref_hs_mol
 Source Type           = specie
 Labels                = Fe...
 Coord                 = [3.7430176, 3.9944543, 13.5094621]...
 Number of Units (Z)   = 1


 Num of Molecules:     = 1
 With Formulae:
    0: H18-C20-N6-S2-Fe ,
--------------------------------
------ SCOPE STATE Object ------
--------------------------------
 Name                  = pbe_opt
 Source Name           = ref_hs_mol
 Source Type           = specie
 Labels                = Fe...
 Coord                 = [3.738143, 3.85153, 13.537435]...
 Number of Units (Z)   = 1


 Num of Molecules:     = 1
 With Formulae:
    0: H18-C20-N6-S2-Fe ,
--------------------------------
------ SCOPE STATE Object ------
--------------------------------
 Name                  = b3lyp_opt
 Source Name           = ref_hs_mol
 Source Type           = specie
 Labels                = Fe...
 Coord                 = [3.774774, 3.779086, 13.494815]...
 Number of Units (Z)   = 1

 Has VNMs              = YES
 Is Minimum            = True
 Is a Transition State = False
 First Freq (cm-1)     = 21.6469

 Num of Molecules:     = 1
 With Formulae:
    0: H18-C20-N6-S2-Fe ]
```

In [6]:
```
## States are accessible directly from the list
initial_state = source.states[0]
## Or through the find_state function
found, initial_state = source.find_state("initial")
## As in other find functions, find_states returns: a boolean, and the actua
print(found)
print(initial_state)
```

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```
True
--------------------------------
------ SCOPE STATE Object ------
--------------------------------
 Name                  = initial
 Source Name           = ref_hs_mol
 Source Type           = specie
 Labels                = Fe...
 Coord                 = [3.7430176, 3.9944543, 13.5094621]...
 Number of Units (Z)   = 1


 Num of Molecules:     = 1
 With Formulae:
    0: H18-C20-N6-S2-Fe
```

In [7]:
```
## This is the "initial" state, and is automatically created when adding a s
## Notice that the coordinates, for example, are just the same than those of
print(np.array_equal(source.coord, initial_state.coord))
```

```
True
```

In [8]:
```
## Computations take input information from states, and bring new output inf
## States store all computations that have modified them:
found, pbe_state = source.find_state("pbe_opt")
print(pbe_state.computations)

## Here, only one computation worked on the "initial" State, and stored its
## Notice Initial State and Final State attributes
```

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```
[----------------------------------------------------
   >>> >>> >>> >>> COMPUTATION
----------------------------------------------------
 Source Type           = specie
 Source sub-Type       = molecule
 Branch Name           = isolated
 Workflow Name         = ref_hs_mol
 Job Name              = opt
 Job Type              = opt
----------------------------------------------------
 Initial State         = initial
 Final State           = pbe_opt
 Comp software         = g16
 Comp index            = 1
 Comp step             = 1
 Comp run_number       = 1
 Comp inp_path         = /Users/sergivela/Documents/SCOPE/Program/Scope_Tuto
rials/Data/1-Tutorial_1/Computations/ABITEM/isolated/ABITEM_ref_hs_mol_opt_r
1.com
 Comp out_path         = /Users/sergivela/Documents/SCOPE/Program/Scope_Tuto
rials/Data/1-Tutorial_1/Computations/ABITEM/isolated/ABITEM_ref_hs_mol_opt_r
1.log
 Comp isregistered     = True
 Comp isgood           = True
 Comp isfinished       = True
 Comp elapsed_time     = 54358.6 seconds
 Has OUTPUT            = NO

]
```

In [9]:
```
## In this Computation, a DFT optimization with PBE was performed.
## Notice that the coordinates of the pbe_opt State are different.
print(source.coord[0])        ## The original coordinates
print(initial_state.coord[0]) ## The initial state coordinates, with the sam
print(pbe_state.coord[0])   ## The final state coordinates, with the optimiz
```

```
[3.7430176, 3.9944543, 13.5094621]
[3.7430176, 3.9944543, 13.5094621]
[3.738143, 3.85153, 13.537435]
```

In [10]:
```
## If multiple computations target the same final state, these keep updating

## For instance, when multiple optimization computations are necessary:
   ## Computation 1: Reads initial state (coordinates "A"), optimization run
   ## Computation 2: Reads set of coordinates "B", continues optimization, a
   ## ... This goes on until the final coordinates are considered good, or t
```

Why are States necessary?

In [11]:
```
## For two reasons:
```

In [12]:
```
# (1) When a computation finishes, the final information may be different th
# For instance, a geometry optimization might break a bond, so the initial m

# Thus, if the user wants to get the molecules out of a STATEs coordinates,
```

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```
new_molec = pbe_state.get_molecules()[0]

## In most cases, the new molecule will have the same chemical composition t
original_molec = source

## Notice that Species can be compared based on some compositional criteria
print(original_molec == new_molec)
```

```
True
```

In [13]:
```
# (2) Because it enables the user to store multiple results of the molecule
# for instance, multiple optimization jobs with different DFT functionals:
found, b3lyp_state = source.find_state("b3lyp_opt")

print(pbe_state.results)    # REMINDER: above, we defined pbe_state, as the
print(b3lyp_state.results)
```

```
{'energy': energy: -3158.67305783 au}
{'energy': energy: -3159.62579289 au}
```

In [14]:
```
# The b3lyp state loaded above contains the information of two computations.
print(len(b3lyp_state.computations))

# Associated with two different jobs, with names:
for comp in b3lyp_state.computations:
    print(comp._job.name)
```

```
2
opt2
freq
```

In [15]:
```
# The second one computes the vibrational normal modes, and stores them:
print(b3lyp_state.VNMs[0])  ## This is the first one
```

```
-----------------------------------
------   SCOPE VNM Object    -----
-----------------------------------
 Index                  = 1
 Freq (cm-1)            = 21.6469
 IR Intensity (KM/Mole) = 1.0847
 Reduced Mass (AMU)     = 14.2624
 Has Mode               = False
```

In [16]:
```
# And this is the resulting IR spectrum
b3lyp_state.plot_ir_spectrum()
```

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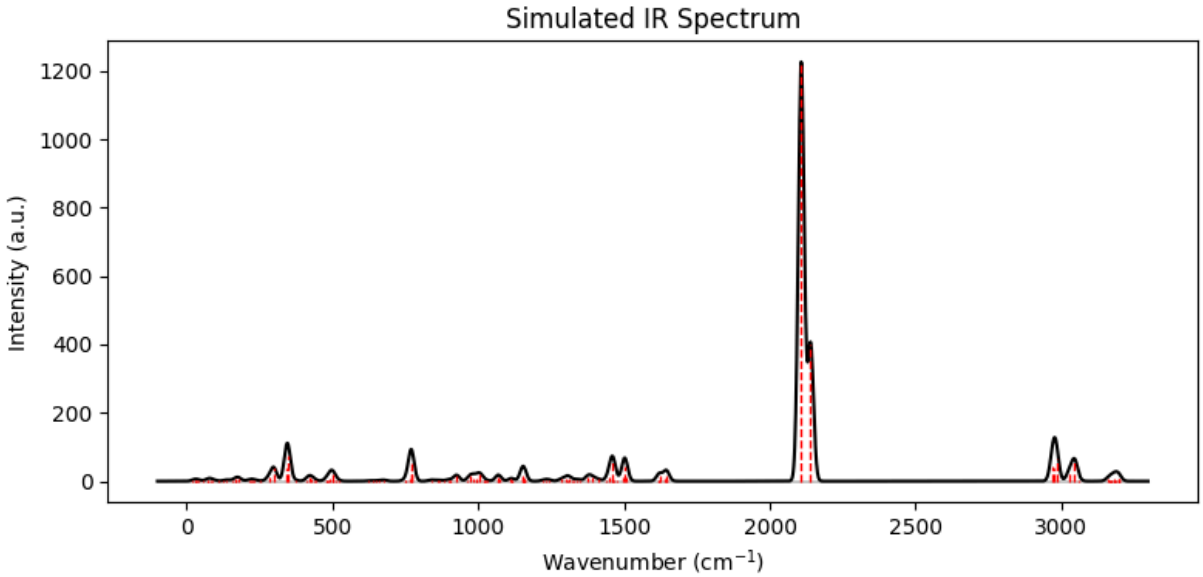

Tools

Sampling Geometries

In [17]:
```
## We implemented in SCOPE some tools that might be useful in computational
```

In [18]:
```
## For instance, the function state.sample_geometries() does a geometry samp
```

In [19]:
```
## By default, the eigenvectors (modes) are not stored in the VNM object. We
comp   = b3lyp_state.computations[-1]                                # Here,
output = comp.create_output()                                        # Here,
vnms   = output.get_vnms(witheigen=True)                             # Here,
b3lyp_state.set_VNMs(vnms)                                           # Here,
```

In [20]:
```
# Here, 10 geometries are generated. The function returns the Q displacement
qdis, geom, ener = b3lyp_state.sample_geometries(10, n_fps_rounds=0, temp=30

# The Sampling is performed using a normal distribution for each mode, with
# Geometries that change the molecular connectivity are discarded, which is
# Large unphysical displacements are also discarded. This breaks the expecte

# The function may return fewer than `n_samples` geometries if the adjacency
# When this happens, a warning is displayed
```

In [21]:
```
## One way to compute how different is one geometry from the original one, i
import numpy as np
maxdisp = 0
for q in qdis:
    if np.linalg.norm(q) > maxdisp: maxdisp = np.linalg.norm(q)
print(np.round(maxdisp,4))
```

```
3.6131
```

In [22]:
```
# Tr[illegible]ase the sampling amplitude (i.e. the distance from the reference g
qdis, geom, ener = b3lyp_state.sample_geometries(10, n_fps_rounds=0, temp=10
```

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```
maxdisp = 0
for q in qdis:
    if np.linalg.norm(q) > maxdisp: maxdisp = np.linalg.norm(q)
print(np.round(maxdisp,4))
```

```
Warning: Reached maximum count of 1000 with only 8/10 samples created.
5.4695
```

In [23]:
```
# A better alternative is to perform different rounds of sampling, and selec
qdis, geom, ener = b3lyp_state.sample_geometries(10, n_aux_geoms=20, n_fps_r

maxdisp = 0
for q in qdis:
    if np.linalg.norm(q) > maxdisp: maxdisp = np.linalg.norm(q)
print(np.round(maxdisp,4))
```

```
Warning: Reached maximum count of 2000 with only 19/20 samples created.
Warning: Reached maximum count of 2000 with only 19/20 samples created.
6.4348
```

In [24]:
```
# If this worked as expected, you should see that the maximum displacement i
# The second method is better for sampling purposes, but it can be slow.
```

How to Parse QC Computations **Manually** and Store Results in a State

---

Normally, SCOPE parses and manages the results of computations that are created by the computational workflow.
However, if you're an advanced user, you might want to update some of these results, or just import some information to SCOPE without having it dealing with the job.

Here, we'll see how we can parse data from a Gaussian16 computation (as an example), and store its results in a state Parts of what is shown here has been discussed above, already. Notably, the Output class

In [25]:
```
## We should have in memory the b3lyp_state we discussed before. Let's selec
comp        = b3lyp_state.computations[-2]

## We get the path:
comp_file   = comp.out_path

## This would be the path of a Gaussian output file that we have. The lines
print(comp_file)
```

```
/Users/sergivela/Documents/SCOPE/Program/Scope_Tutorials/Data/1-Tutorial_1/C
omputations/ABITEM/isolated/ABITEM_ref_hs_mol_opt2_r1.log
```

In [26]:
```
## Parsing computations manually, implies creating an Output object of the c
lines = scope.parse_general.read_lines_file(comp_file)      # First, we rea
```

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```
output = scope.software.gaussian.G16_output(lines)          # Then, we crea
```

```
G16_OUTPUT: output created without Computation.
G16_OUTPUT: Please specify comp_type doing set_comp_type()
G16_OUTPUT: Options are: 'scf', 'opt', 'freq', 'td' and 'tda'
```

In [27]:
```
## Since we're doing this manually, the Output is not associated to a Comput
output.set_comp_type('opt')
```

In [28]:
```
## From here, you can extract energies ...
energy = output.get_last_energy()
print(energy)
```

```
-3159.62579289
```

In [29]:
```
## ... and final geometry
labels, coords = output.get_last_geometry(debug=1)
print(labels[0], coords[0])
```

```
Fe [3.774774, 3.779086, 13.494815]
```

In [30]:
```
## You can use this data to update the state, if you want. For instance:
b3lyp_state.set_energy(energy, 'au')
b3lyp_state.set_geometry(labels, coords)
```

In [31]:
```
## The functions that take care of that in SCOPE are in scope.register_data.
```

In [32]:
```
## Remember that, if you update the systems, sources, or states information
## Just do:
#sys.save() to save in the default path = sys.system_file
## or
#sys.save("whatever_path") if you want to specify the actual path
```

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